Yassine Ben Maktouf

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Junior Actuary and Data Science Engineer

Experience

- **Actuarial Data Scientist (End-of-Studies Internship) | Vermeg, Tunis** *Feb 2026 - Jul 2026*
  - Automated the configuration of PERI retirement-savings products in Solife, Vermeg's life-insurance and pensions platform, from specification documents: an LLM extracts intent, then a deterministic, auditable engine (lxml/XPath, layered safety guards) applies each change to regulated XML and Groovy business rules.
  - Built a hybrid RAG assistant over an actuarial and insurance corpus (all-MiniLM dense retrieval + BM25 with Reciprocal Rank Fusion); benchmarked embedding models and retrieval strategies on MRR, Recall and NDCG to select the production configuration.
  - Compared fine-tuned XLM-RoBERTa with LLMs for named-entity extraction, audited answer faithfulness with NLI, and applied grammar-constrained decoding to make extraction reliable; delivered a multi-instance React + FastAPI platform with JWT authentication and per-instance retrieval isolation.
- **Data Science Intern | Les Laboratoires MédiS, Nabeul** *Jun 2025 - Aug 2025*
  - Developed a sales forecasting system, comparing machine learning and deep learning models and tuning the best performer.
  - Built an automated data pipeline integrated with SQL Server.
  - Delivered a web application with interactive reporting and visualisations to support business decisions.
- **Software Engineering Intern | STIP, Tunis** *Jun 2024 - Jul 2024*
  - Built a SQL Server database, Node.js REST APIs and a React/Bootstrap interface to manage employees, phone numbers and contracts, with automatic number assignment, change tracking and role-based access control.

Key Projects

- **Actuarial Portfolio Heterogeneity Study (R):** Modelled the mortality of French smokers and non-smokers (1950 cohort) with the Lee-Carter model, projected future mortality trends, and designed fair annuity pricing for each group.
- **AI Platform for Medical Insurance Documents:** Processed handwritten French/Arabic medical documents with YOLO classification, fine-tuned OCR, zero-shot layout detection and a RAG-based LLM; added signature-based fraud detection (Siamese network) and reimbursement estimation. Deployed with FastAPI, Docker and Kafka; presented at ESPRIT's 12th Bal des Projets (2025).
- **Customer Churn Prediction:** Telecom churn pipeline with Random Forest and Decision Tree on SMOTE-balanced data, plus PCA and K-Means to identify churn drivers; deployed through a REST API. github.com/YassineBenMaktouf/Customer-Churn
- **Audio Emotion Recognition:** Combined a diffusion model with a ResNet-50 classifier for speech emotion recognition on mel-spectrograms, evaluated on the EmoDB and RAVDESS datasets.
- **Hackathon Projects:** Imaginify AI, an AI story-generation web app (1st place, Special AI Models Hack, lablab.ai, 2024); FinAssist, an AI personal-finance assistant that analyses Excel data (2nd place, ChatHack, lablab.ai, 2023); NASA International Space Apps Challenge (2024).

Education

- **Master's in Actuarial Science (double degree) | IRA**, Le Mans University, France *2024 - 2026*
  Courses: life and non-life actuarial science, risk theory, reinsurance and extreme risks, insurance regulation and prudential standards, stochastic models, Monte Carlo methods.
- **Engineering Degree in Computer Science, Data Science Specialty | ESPRIT**, Tunisia *2023 - 2026*
  Courses: advanced machine learning and deep learning, time series, big data, MLOps and DevOps, probability and statistics, operations research.
- **Preparatory Classes in Mathematics and Physics | IPEIT**, Tunisia *2021 - 2023*
  Courses: analysis, linear algebra, probability, physics.
- **Baccalaureate in Mathematics | Lycée Ibnou Rachik**, Tunisia *2021*

Skills & Certifications

Actuarial & risk: mortality modelling (Lee-Carter), annuity pricing, risk theory, stochastic models, Monte Carlo simulation
Programming: Python, R, SQL, Java, JavaScript, HTML, CSS, C, C#
Machine learning & AI: scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face Transformers, YOLO, NLP, LLM, RAG
Frameworks & MLOps: FastAPI, Flask, React, Spring Boot, .NET, Docker, Kafka, MLflow, GitHub Actions, CI/CD
Databases: SQL Server, MongoDB, MySQL, SQLite, vector databases
Certifications: IBM: Machine Learning with Python, Deep Learning & Neural Networks with Keras, Introduction to Artificial Intelligence; NVIDIA: Applications of AI for Anomaly Detection, Building Transformer-Based NLP Applications, Evaluation and Light Customization of LLMs; AWS Academy Cloud Foundations; Cisco CCNA: Introduction to Networks
Activities: General Secretary of the ATAST club at ESPRIT (2023 - 2024), club ranked 2nd at the annual BTech competition; Member of the IEEE RAS (Robotics and Automation Society) student chapter at ESPRIT (2023 - 2024)
Languages: Arabic (native), English (bilingual), French (professional)